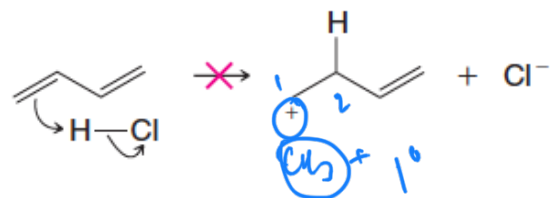
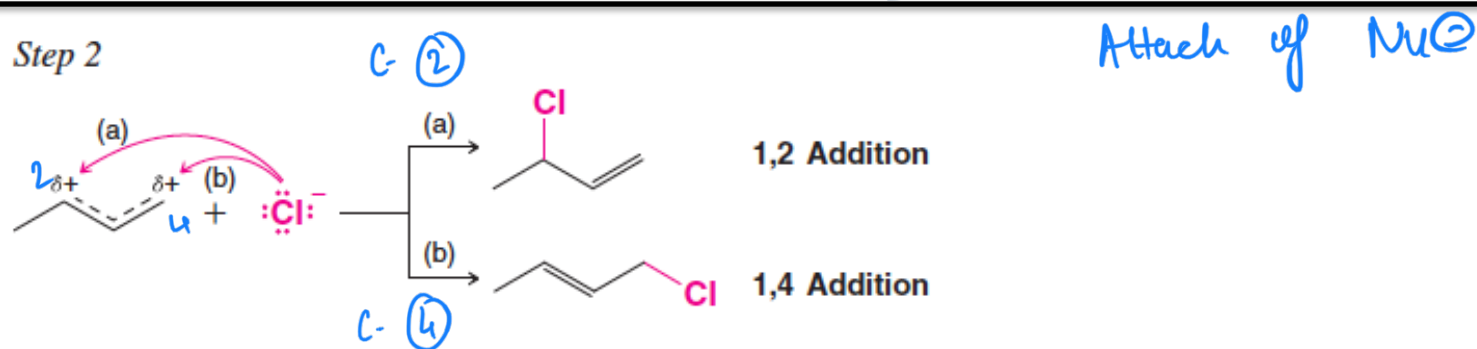
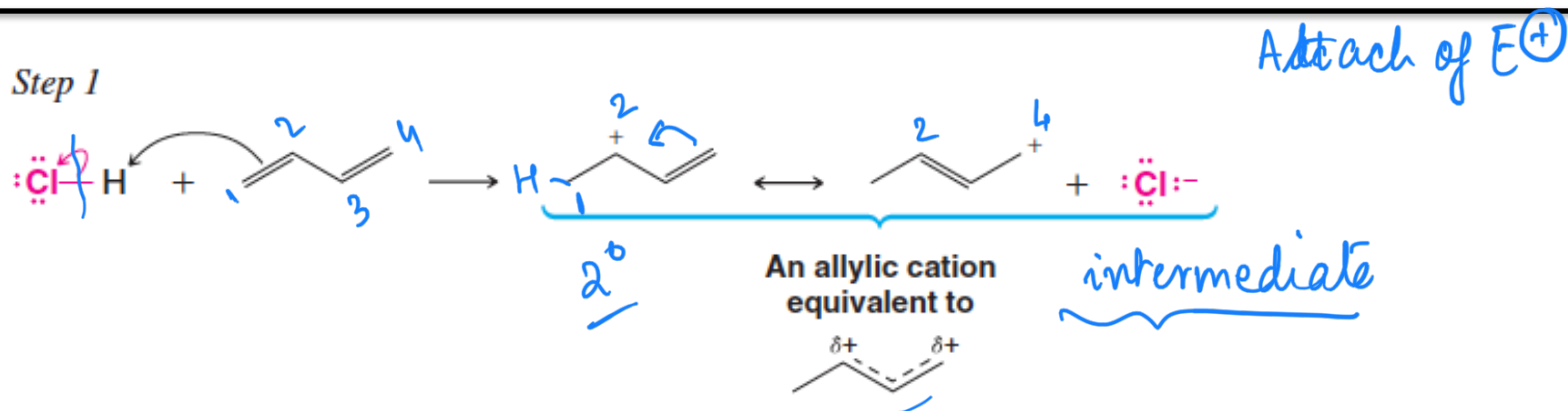
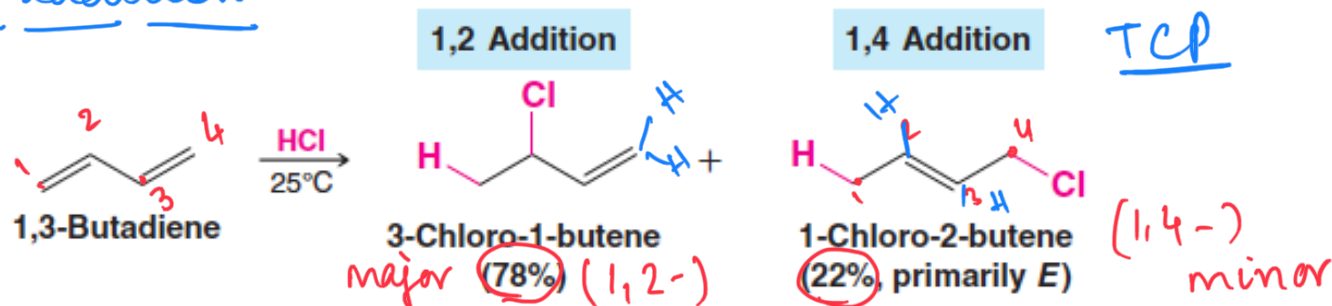
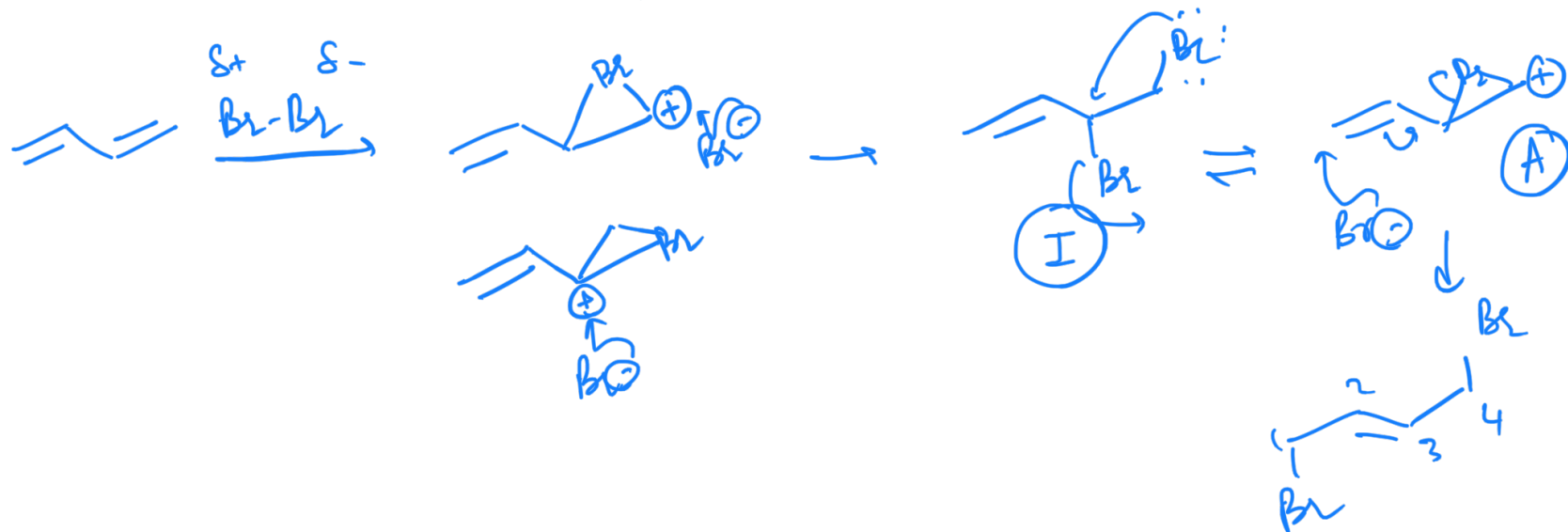
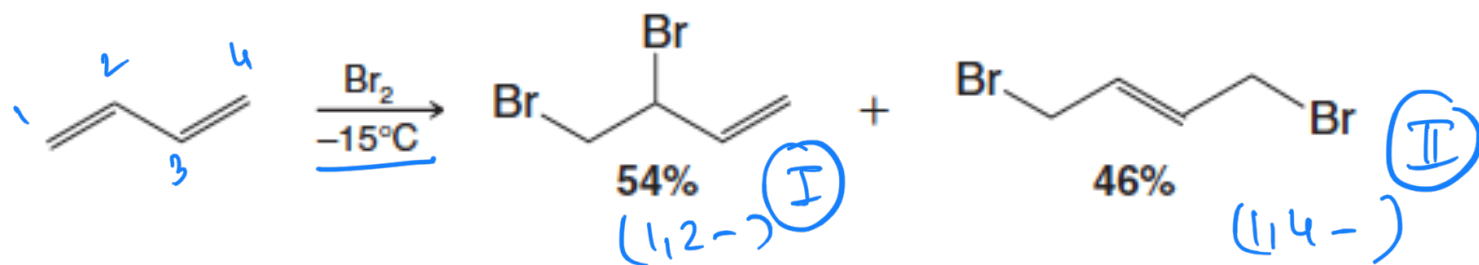
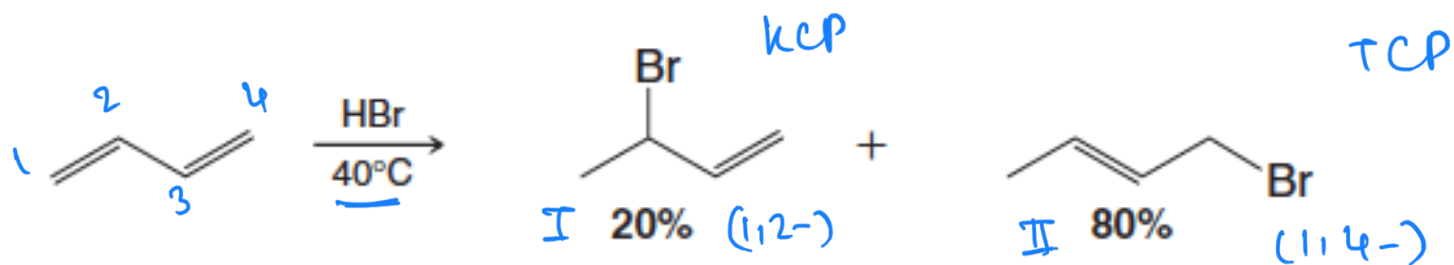


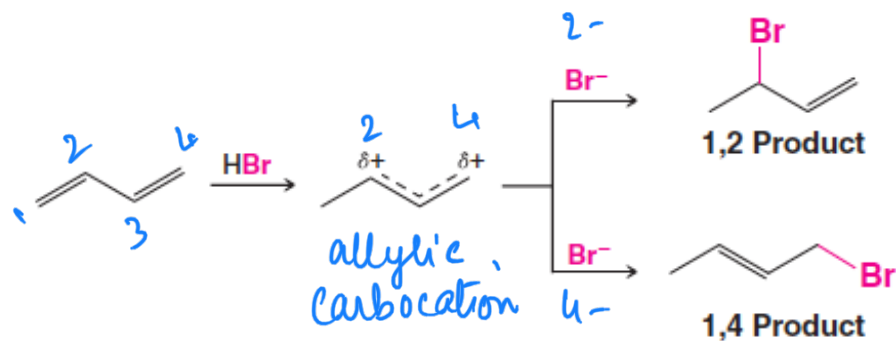
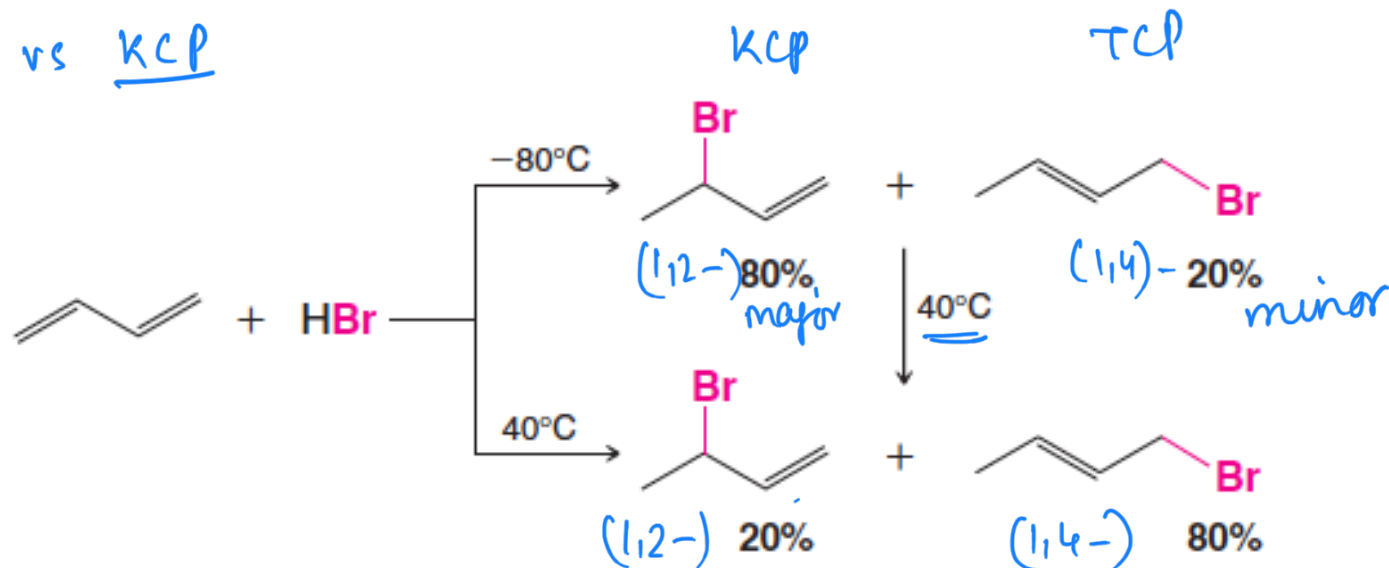
② Conjugate addition



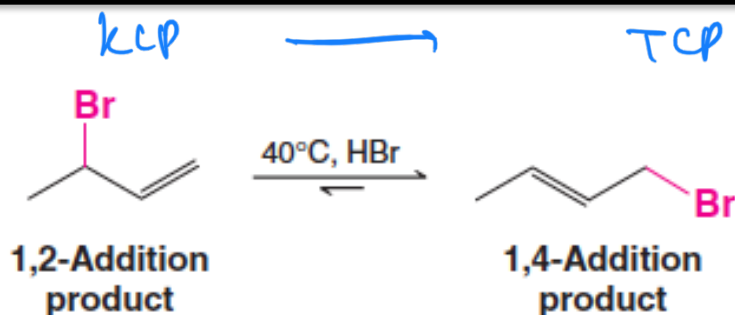
Addition of the electrophile in this fashion does not lead to an allylic (resonance-stabilized) carbocation.



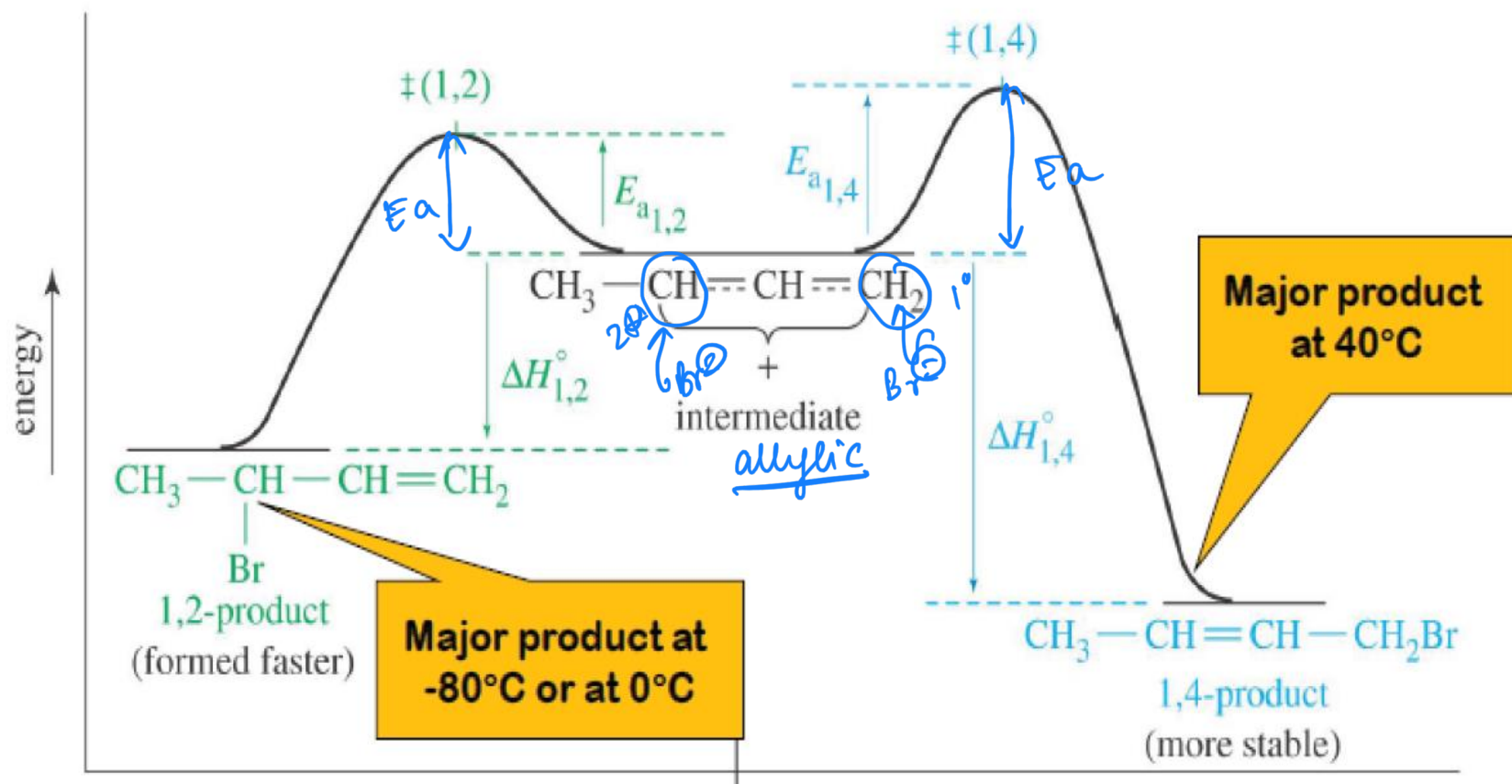
TCP vs KCP



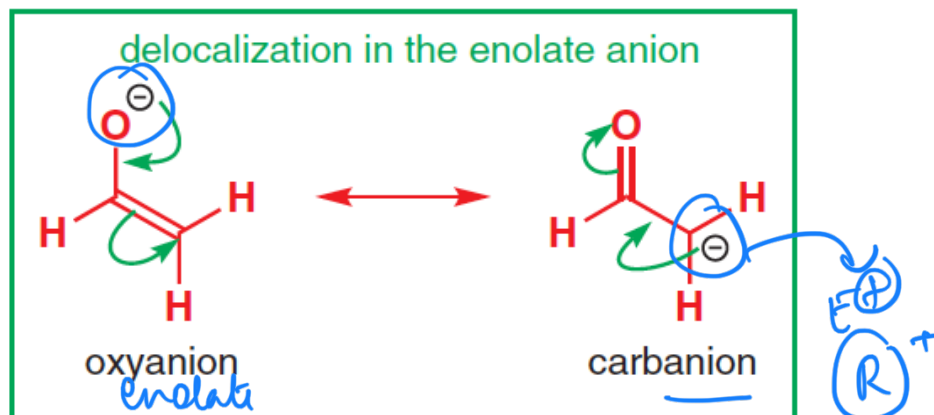
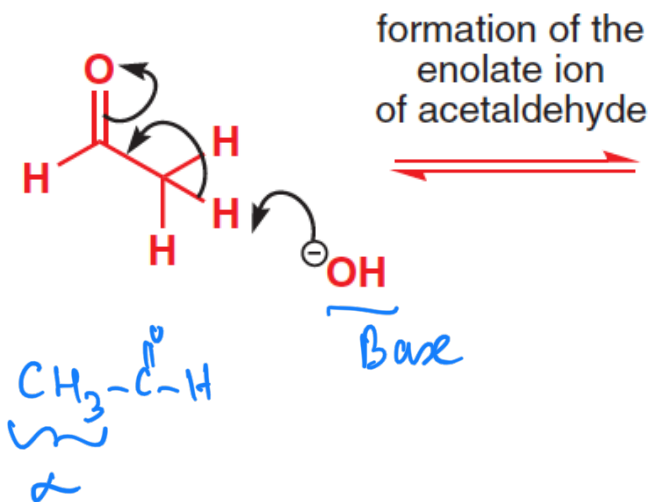
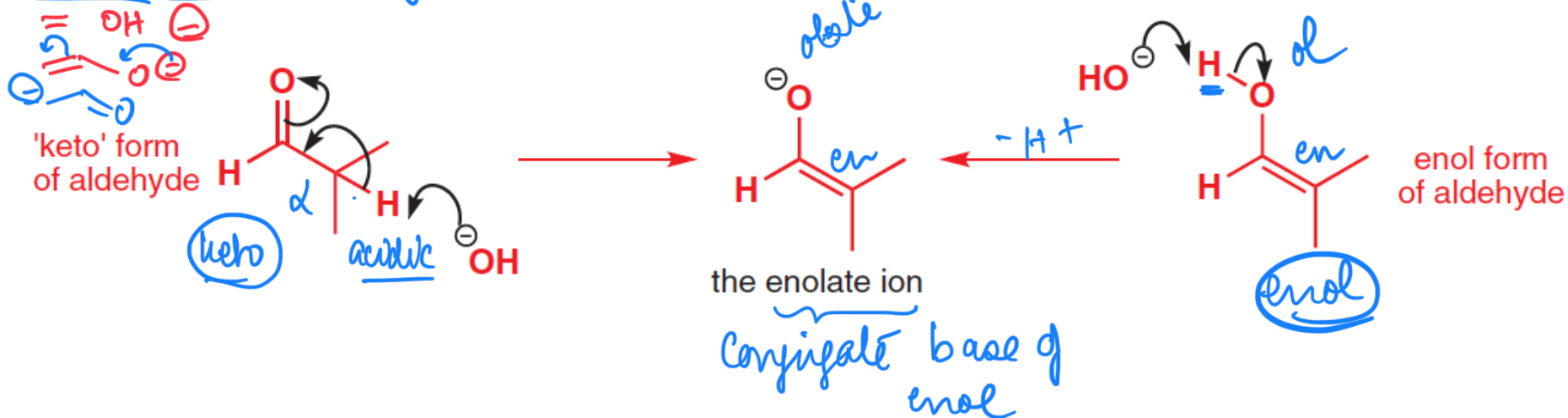
The reaction of the bromide anion with the allylic cation determines the regioselectivity of the reaction.



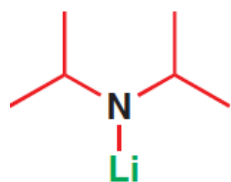
- * Raising temp
- * extending reaction time
- * Common Intermediate



Enolate chemistry:

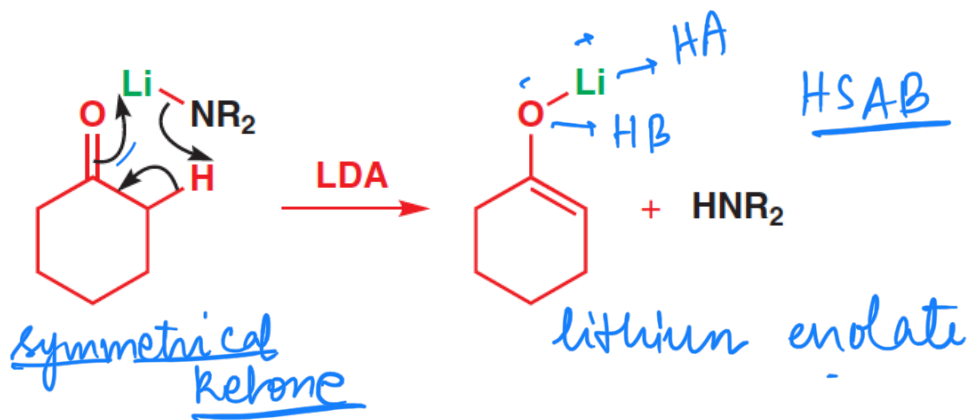


Generation of enolates

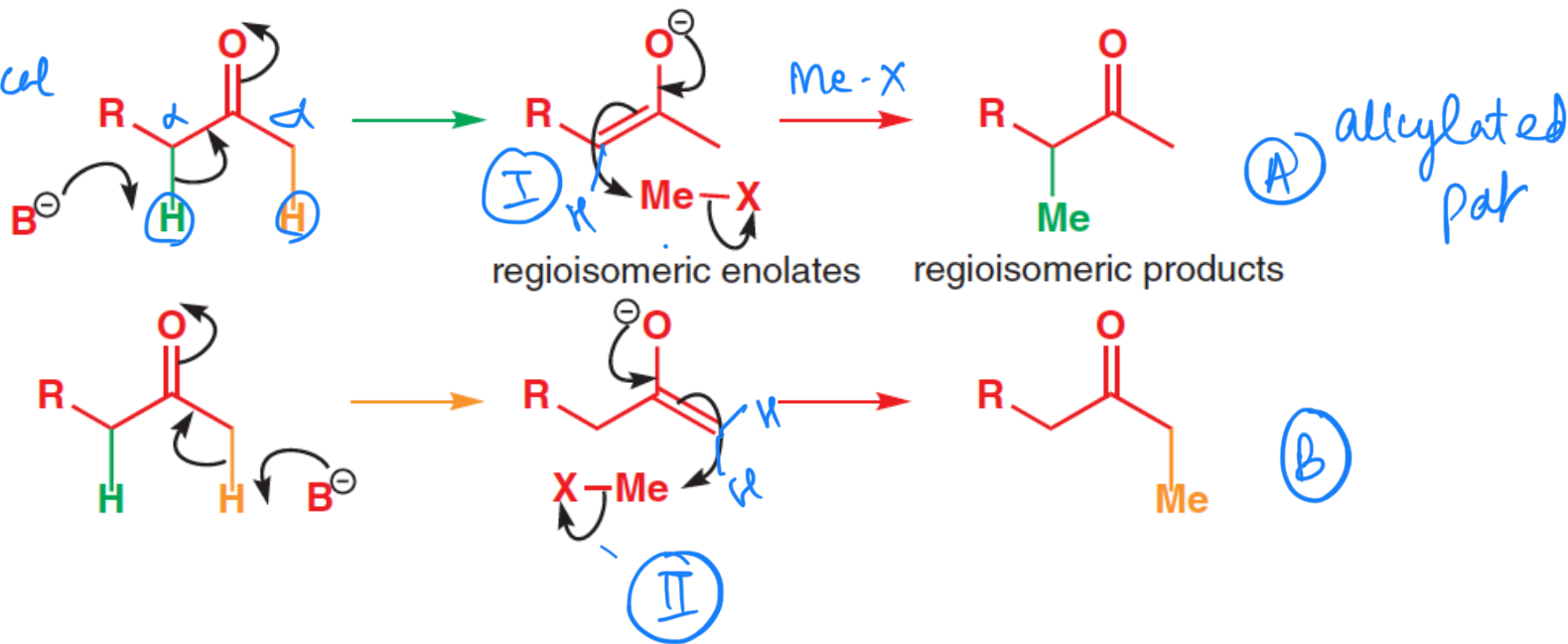


LDA =
lithium
disopropyl
amide
Base

Bulky, Organic,
Non-nucleophilic base



unsymmetrical
ketone



(I) → Thermodynamically stable enolate
(II) → Kinetic enolate

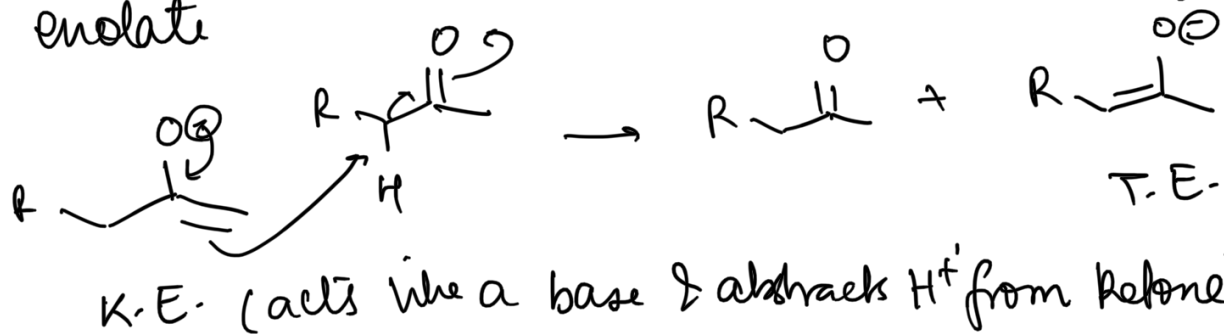
Thermodynamic enolate

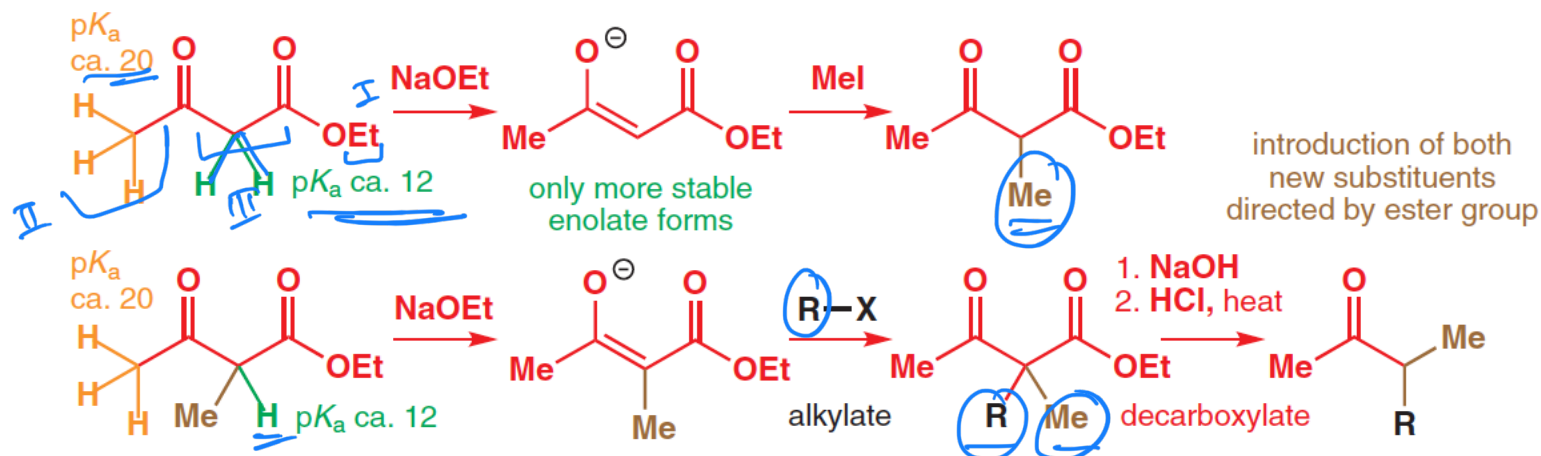
- * more substituted
- * more stable
- * favoured by
 - excess ketone
 - high temp.
 - long reaction time

Kinetic enolate

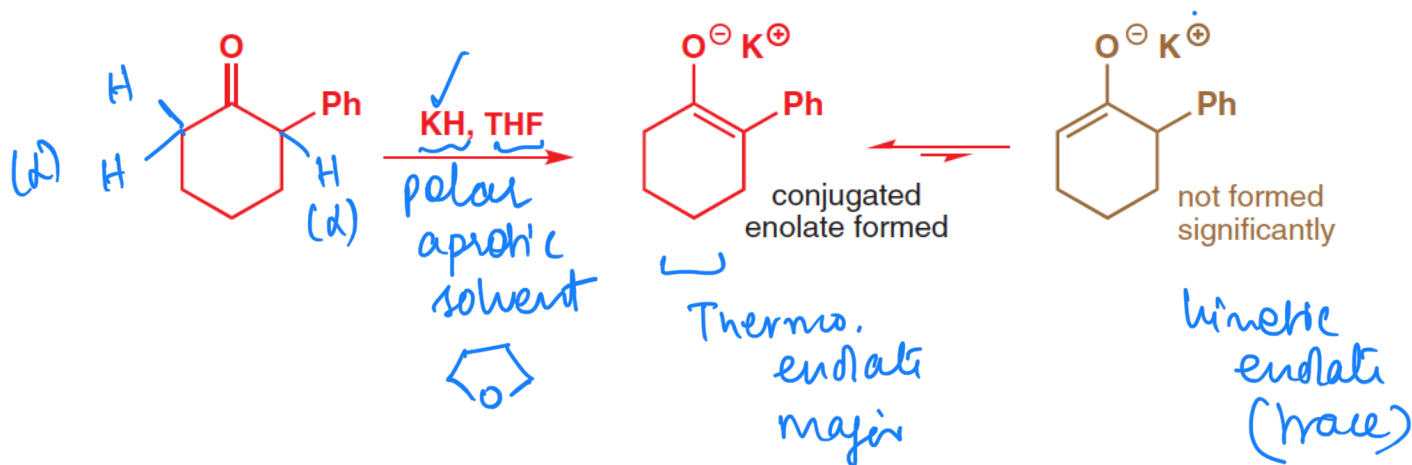
- * less substituted
- * less stable
- * favoured by
 - strong, hindered base
 - low temp.
 - short reaction time

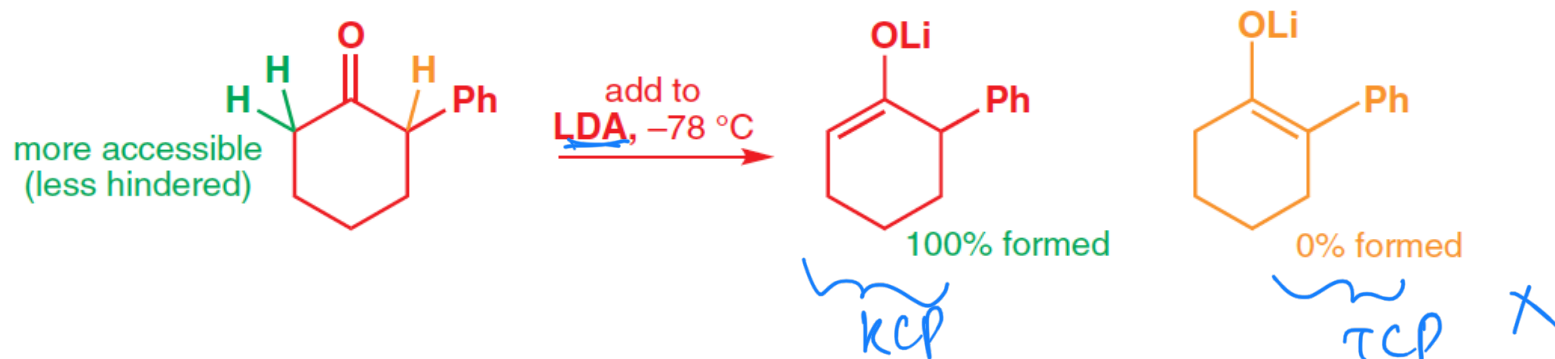
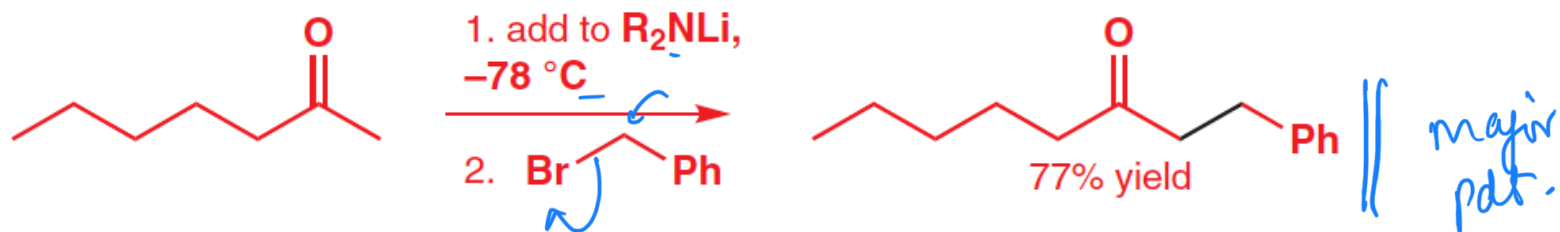
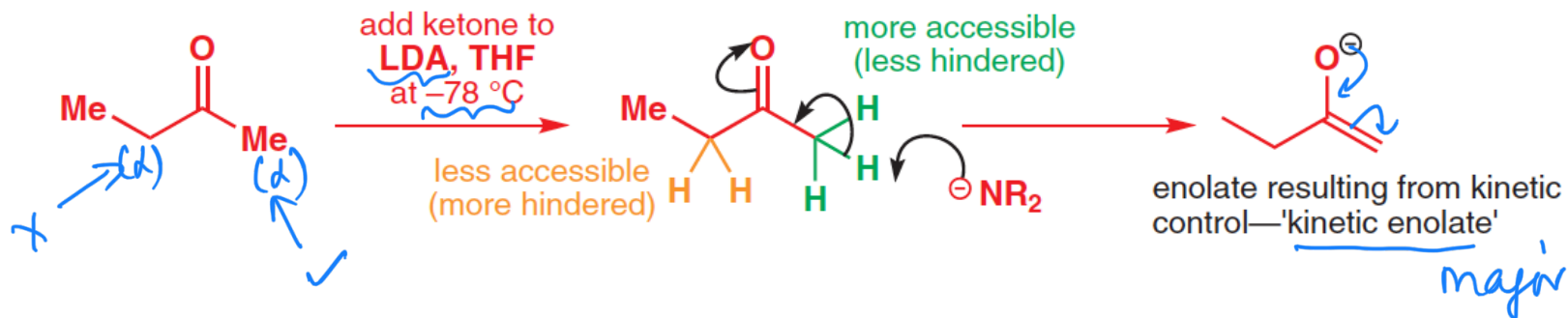
If excess ketone, kinetic enolate can change to thermodynamic enolate





EAA: ethyl acetoacetate

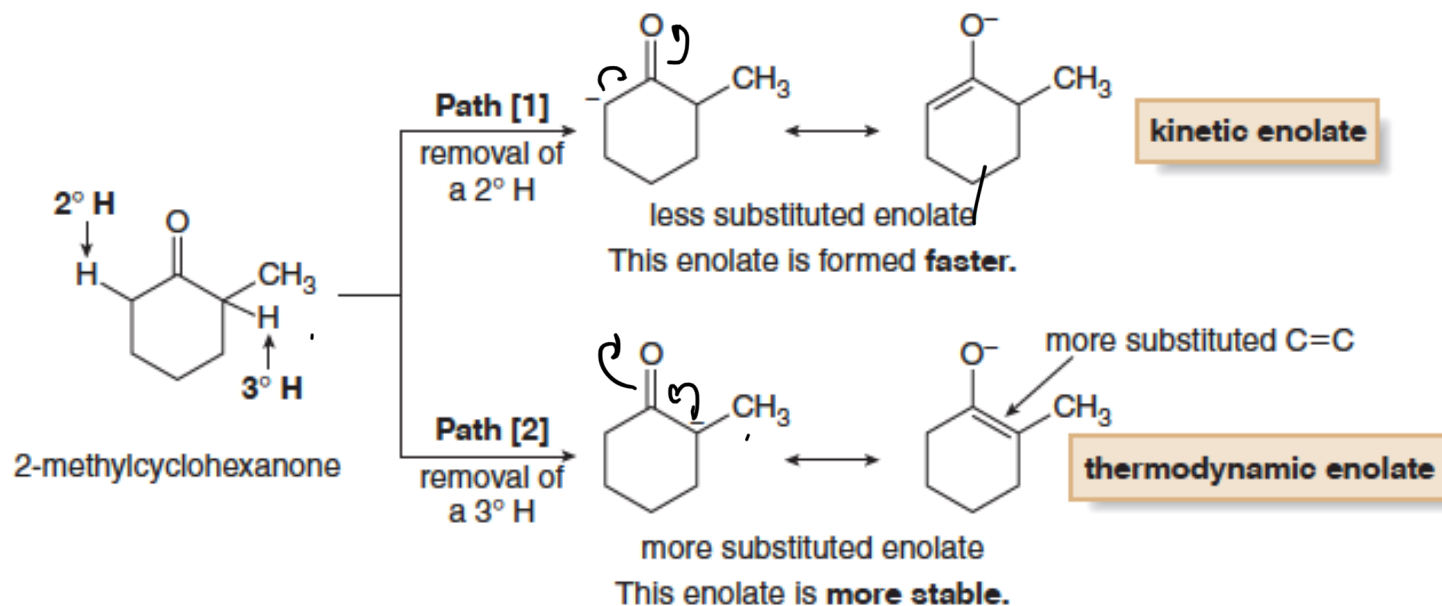
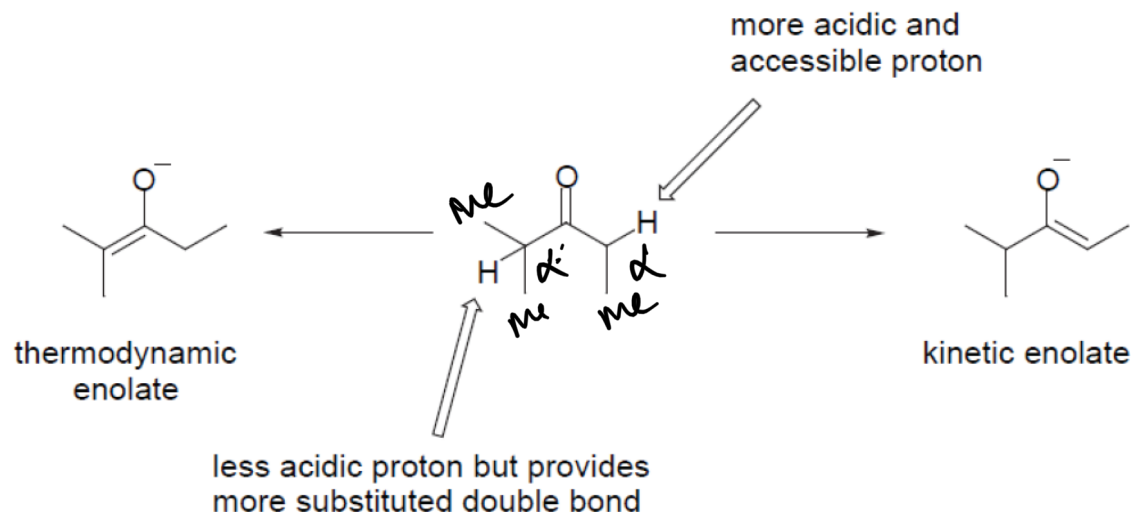


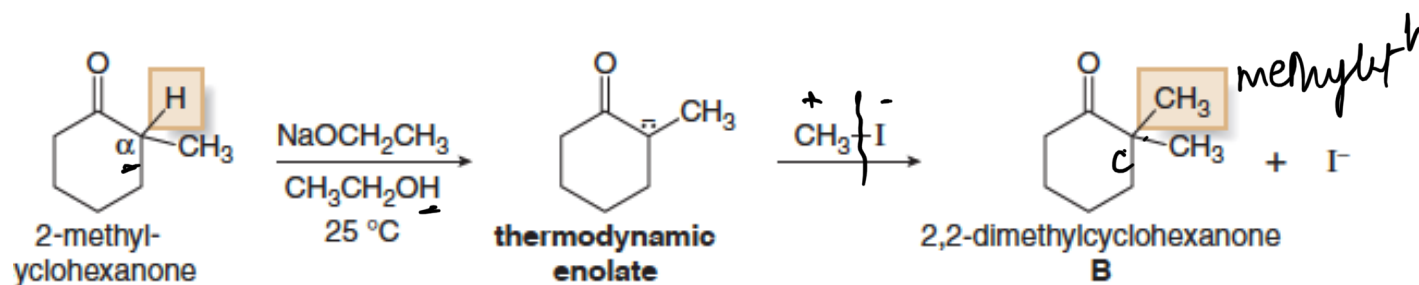
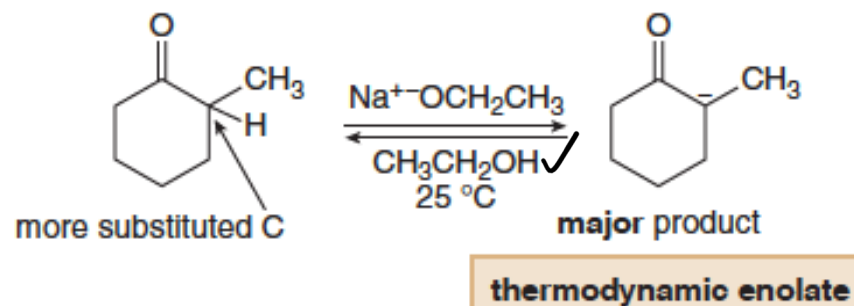
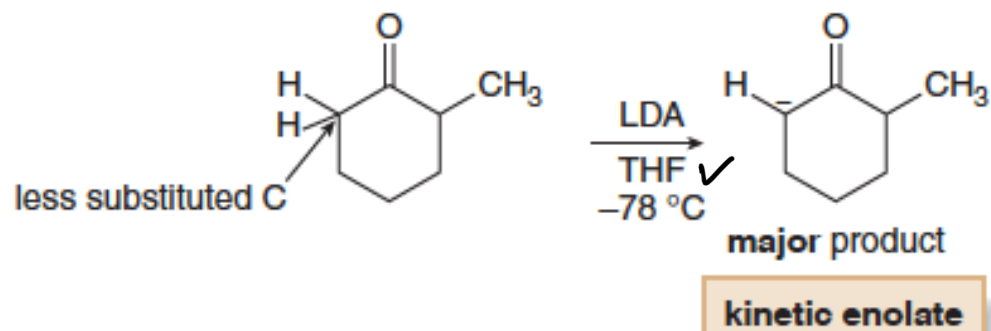


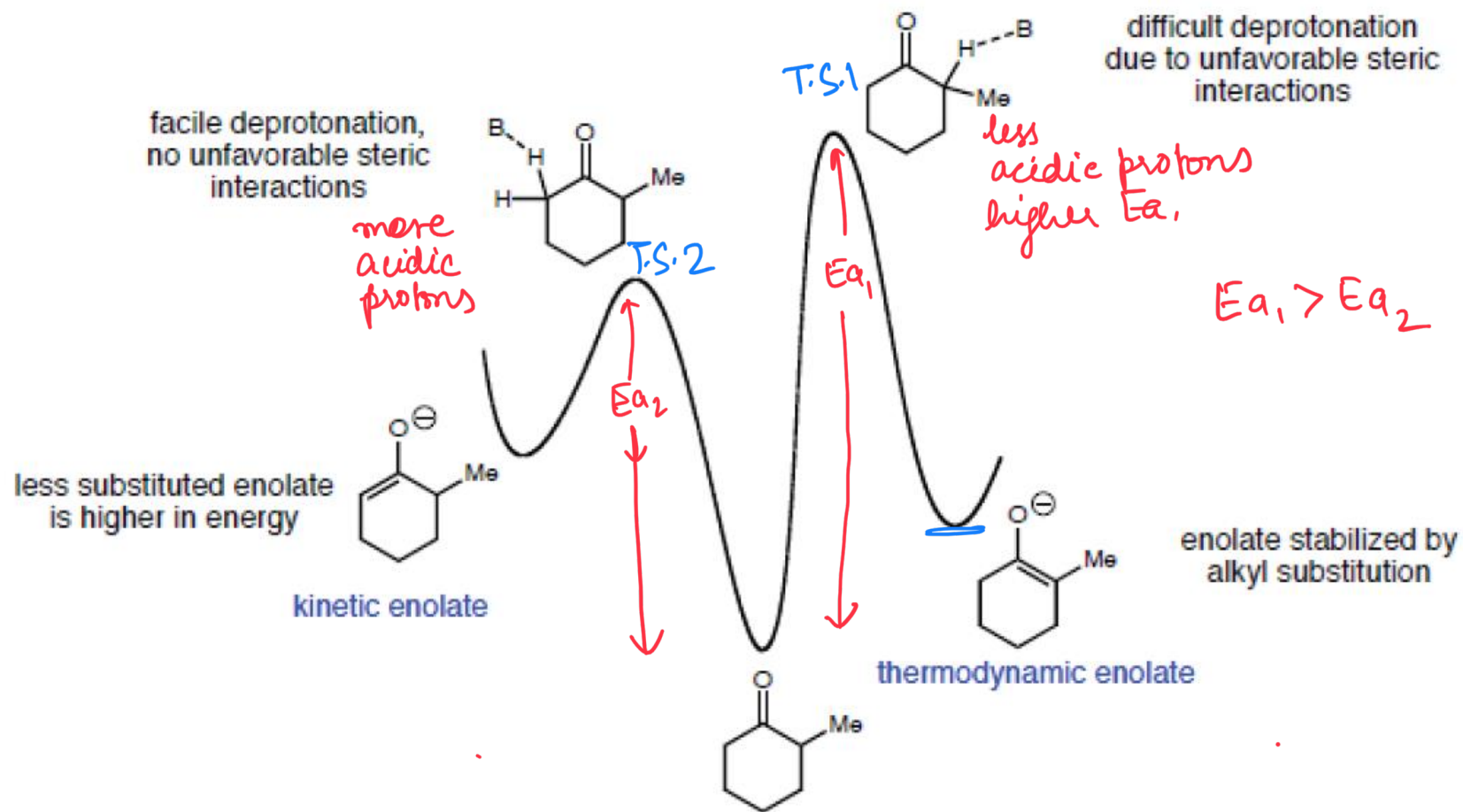
Kinetically controlled enolates : requirements

1. excess base
2. Base to be bulky, strong, non-nucleophilic
3. Polar aprotic solvent
4. low temp.
5. short reaction time

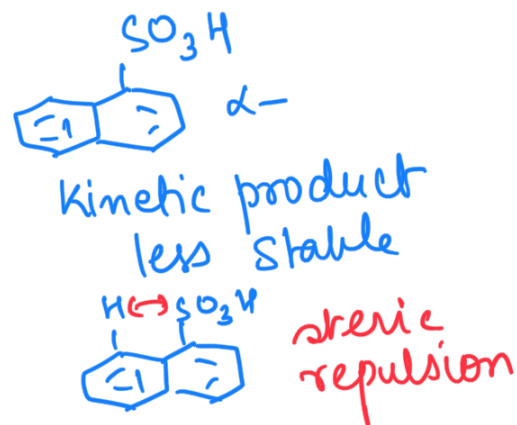
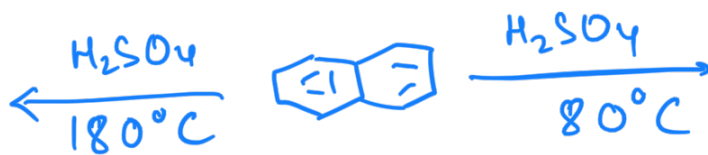
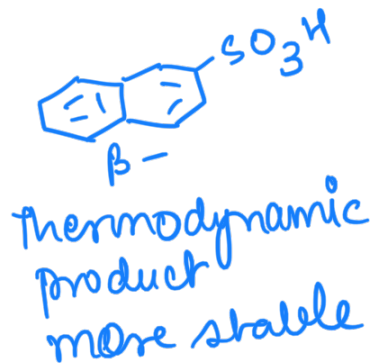
The selective formation of enolates



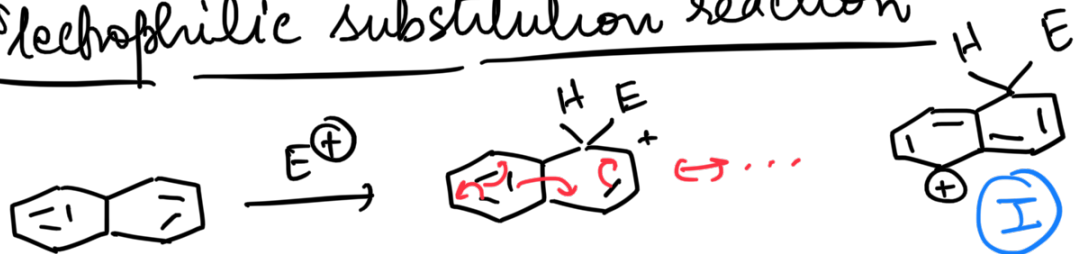




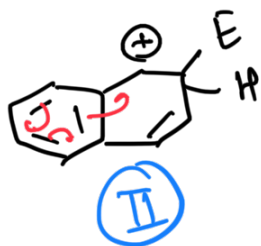
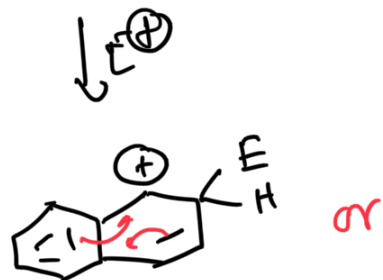
④ sulphonation of naphthalene



Electrophilic substitution reaction

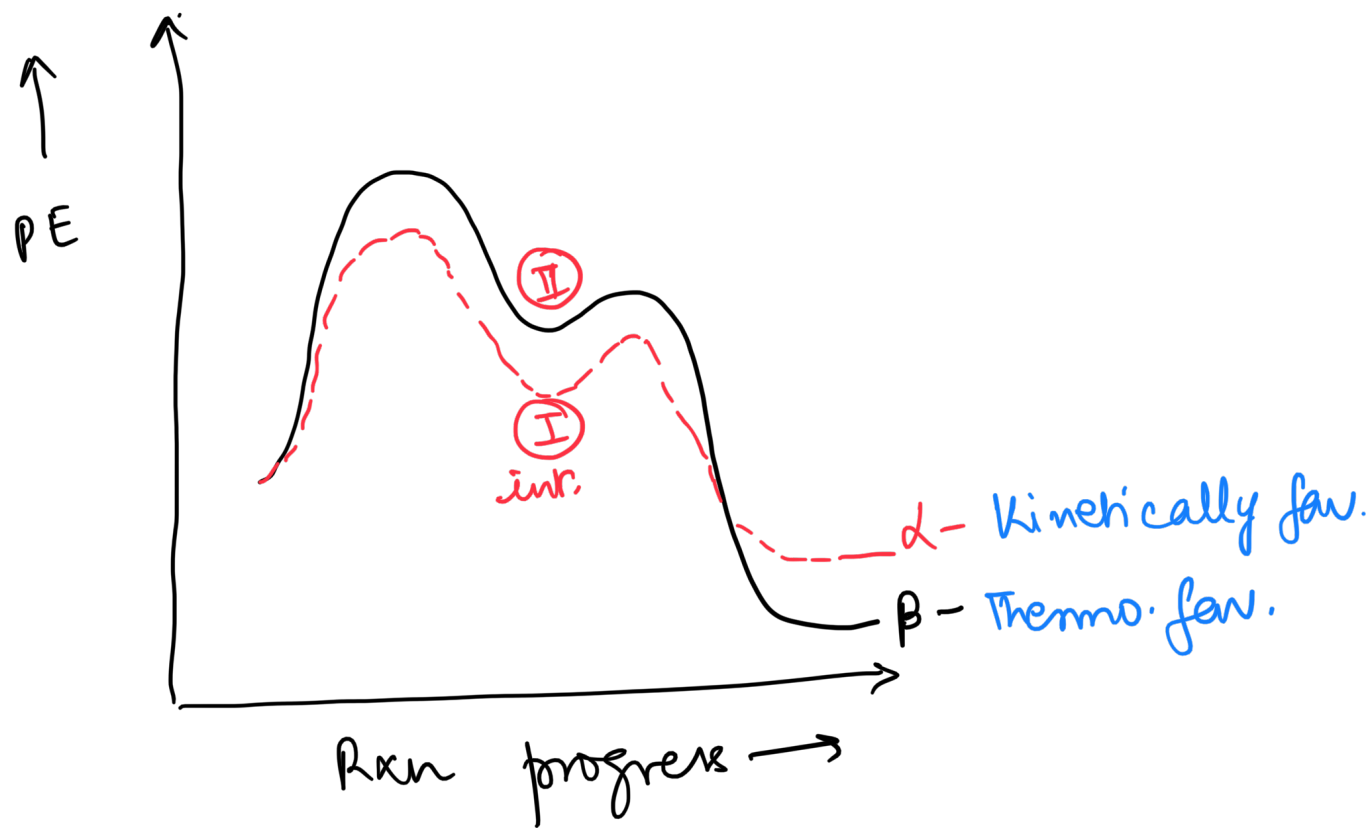


more resonance stabilized.
extended conjugation



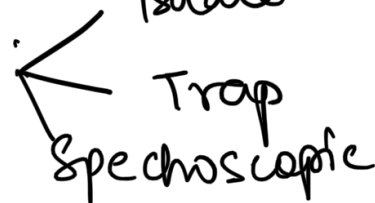
less resonance stabilized
cross-conjugation

Intermediate **I** easily formed → lower E_a
 → gives α - as kinetic product



How to determine a reaction mechanism?

Strategies adopted

- 1) Nature of products
- 2) Kinetics of reaction
- 3) Kinetic isotope effect
- 4) Isotopic labelling
- 5) Intermediate detection 
 - Isolate
 - Trap
 - Spectroscopicin-situ method
- 6) Stereochemistry of reaction

① Nature of products

